

Supplementary Information

Nonlinear magnon dynamics in Co-doped YIG: four operating regimes from cryogenic Fock confinement to room-temperature THz phase encoding

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This document provides the numerical protocols, parameter tables, derivations, and reproducibility checklist for the main text. Section S8 documents the completed finite-temperature frequency scan at $T = 310$ K.

Supplement 1. Numerical models, parameters, and reproducibility

S1.1 Common open-system convention

The full quantum calculations use a single bosonic mode with annihilation operator $\hat{a}$, number operator $n = \hat{a}^\dagger \hat{a}$, and a zero-temperature single-quantum loss channel. The master equation is

$$d\rho/dt = -i[H(t), \rho] + \kappa D[\hat{a}]\rho, \quad D[\hat{a}]\rho = \hat{a}\rho\hat{a}^\dagger - (1/2)\{\hat{a}^\dagger\hat{a}, \rho\}.$$

The damping convention is $\kappa = 2\alpha\Omega$. With this convention the coherent amplitude decays at the Gilbert rate $\alpha\Omega$ because the dissipator contributes $-(\kappa/2)\langle\hat{a}\rangle$ to $d\langle\hat{a}\rangle/dt$.

The calculation scripts use $\Omega = 1$, $U = 0.55$, $\eta = 1.2$, $\alpha = 0.01$, $\kappa = 0.02$, a 50-state Fock basis, normalized time Ωt from 0 to 100, and 3001 requested output times. QuTiP integrates adaptively between these output times.

S1.2 Hamiltonian conventions

The matched audit calculations use the nonlinear Hamiltonian $H_0 = \Omega \hat{a}^\dagger \hat{a} + (U\Omega/2) \hat{a}^{\dagger 2} \hat{a}^2$.

Here U is the dimensionless Kerr nonlinearity coefficient; the corresponding nonlinear frequency scale is $K = U\Omega$. Two pair-drive protocols are then compared:

- Static effective pair coupling: $H_{\text{drive}} = (\eta\Omega/4)(\hat{a}^{\dagger 2} + \hat{a}^2)$.
- Explicit resonant modulation: $H_{\text{drive}}(t) = [\eta\Omega \sin(2\Omega t)/2](\hat{a}^{\dagger 2} + \hat{a}^2)$.

Important limitation. The legacy Kerr-coefficient scan used for the manuscript's earlier Figure 3 was generated with a different nonlinear normalization. A term $U\eta^2$ is not numerically identical to $(U\Omega/2)\hat{a}^{\dagger 2}\hat{a}^2$. Since $\hat{a}^{\dagger 2}\hat{a}^2 = n(n-1)$, changing conventions also changes the effective linear frequency and the coefficient multiplying the anharmonicity. The symbol U must therefore not be transferred between those calculations without an explicit normalization conversion. For the two conventions actually used in this work the conversion is a factor of two: with $\Omega = 1$ the matched-audit term $(U\Omega/2)\hat{a}^{\dagger 2}\hat{a}^2$ equals $(U/2)\hat{a}^{\dagger 2}\hat{a}^2$, so a nominal $U = 0.55$ in Supplements 3–5 carries half the anharmonicity of a nominal $U = 0.55$ in the frequency scan of Supplement 8. The two sets are therefore reported separately throughout and are never combined in a single table.

S1.3 Observables

Quantity	Definition	Interpretation
$ v ^2$	Reduced Bogoliubov occupation	Mean occupation estimate, not P_1
$\langle n \rangle$	$\text{Tr}(\rho \hat{a}^\dagger \hat{a})$	Full quantum mean occupation
P_j	$\langle j \rho j \rangle$	Probability of exactly j quanta
$\langle \hat{a} \hat{a} \rangle$	$\text{Tr}(\rho \hat{a} \hat{a})$	Anomalous pair coherence
$\langle (-1)^n \rangle$	Parity expectation	Even/odd-sector imbalance

S1.4 Numerical convergence and provenance

Convergence statements are supported only when directly tied to an archived script and verified output file. Claims not documented in the archive — including any yield, throughput, or Monte-Carlo figures — are not part of the validated results of this paper.

The available 40- and 100-level legacy calculations demonstrate that the upper Fock cutoff was not populated for the stated static effective-drive cases. This is a basis-convergence check; it is not evidence that the physical state is an exact two-level state.

Supplement 2. Relation between the Lindblad and Bogoliubov descriptions

S2.1 Exact linear structure and noise term

For $U = 0$, the Hamiltonian is quadratic and the Heisenberg-Langevin equation has the standard form

$$d\hat{a}/dt = -i[\hat{a}, H(t)] - (\kappa/2)\hat{a} + \sqrt{\kappa} \hat{a}_{\text{in}}(t).$$

The input-noise term is required in the open-system operator equation. A deterministic reduced equation that keeps only the damping term does not preserve the canonical hyperbolic norm of a unitary Bogoliubov transformation.

S2.2 Mean-field nonlinear closure

The quartic Kerr interaction generates higher operator moments, so the dynamics do not close exactly on two complex amplitudes $u(t)$ and $v(t)$. The reduced equations used in the audit replace the operator interaction by a state-dependent frequency shift $\Delta U = U\Omega|v|^2$. This is a Hartree-type closure and is an approximation, not an operator identity.

The matched reduced equations are integrated with the same resonant modulation $G(t) = \eta\Omega \sin(2\Omega t)$ as the corresponding Lindblad calculation. Agreement should therefore be assessed for mean occupation and phase trends, not for the complete Fock distribution or all higher moments.

S2.3 Damped norm diagnostic

For the deterministic reduced equations with equal damping $-\alpha\Omega u$ and $-\alpha\Omega v$, direct differentiation gives

$$d/dt (|u|^2 - |v|^2) = -2\alpha\Omega (|u|^2 - |v|^2),$$

and hence $|u|^2 - |v|^2 = e^{-2\alpha\Omega t}$ for $u(0)=1$ and $v(0)=0$. At $\alpha = 0.01$ and $\Omega t = 100$ this equals $e^{-2\alpha\Omega t} = e^{-2} = 0.1353352832$. The numerical audit reproduces this value with a maximum corrected-norm error below 8×10^{-9} . This diagnostic confirms the integrator; it must not be interpreted as a transition from a hyperbolic to a Euclidean quantum-state normalization.

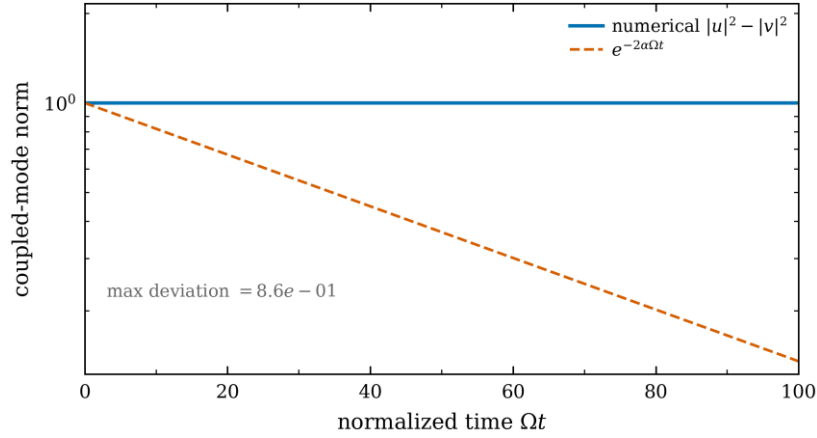


Figure S1. Damped coupled-mode norm diagnostic. The decaying curve follows $\exp(-2\alpha\Omega t)$; $|u|^2 + |v|^2$ is shown only as a diagnostic and is not a state normalization.

Supplement 3. Drive-protocol audit

The purpose of this audit is to isolate the effect of the pair-drive implementation while holding the matched Kerr convention, damping convention, basis size, initial state, and time interval fixed.

Protocol	Late mean $\langle n \rangle$	Late mean P_0	Late mean P_1	Late mean $P(n \geq 2)$	Final parity
Static effective	0.11696	0.91857	0.04952	0.03192	0.89352
Resonant modulation	1.21332	0.31152	0.33347	0.35500	0.15432

The resonantly modulated protocol produces a late-time mean occupation approximately 10.4 times larger than the matched static effective protocol. Thus nominally identical labels (U, η, α) do not identify the same physical calculation when the drive implementation differs.

The matched static audit gives a late-time mean occupation of 0.11696, whereas the legacy Figure 3 calculation reported a final occupation of 0.0404. The difference is not attributed solely to the drive: the legacy calculation also used a different Kerr normalization. These results must remain separately labelled.

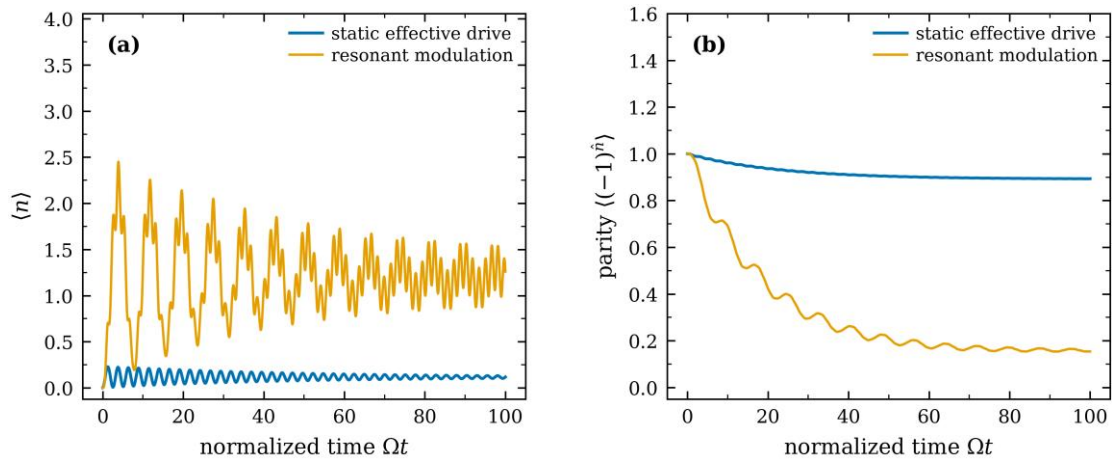


Figure S2. Protocol audit of the mean occupation. The static effective and explicitly resonant pair drives produce distinct bounded occupation scales under the matched Kerr convention.

Supplement 4. Matched Bogoliubov–Lindblad bridge

Under the resonantly modulated protocol at $U = 0.55$, $\eta = 1.2$, $\alpha = 0.01$ and $\kappa = 2\alpha\Omega$, the reduced calculation gives a late-time mean $|v|^2 = 1.11640$ with standard deviation 0.49405. The Lindblad calculation gives a late-time mean $\langle n \rangle = 1.21332$ with standard deviation 0.22237. Their ratio is 0.9201.

The approximately 8% difference in the time-averaged occupation confirms quantitative consistency between the reduced and full Lindblad descriptions at the level expected from the mean-field closure. It does not imply agreement of the instantaneous trajectories: the previously recorded maximum pointwise difference was 2.56 and the RMS difference was 0.72.

The Lindblad state is not a one-quantum Fock state. At the representative resonant point the late-time populations are broadly distributed, with mean $P_0 = 0.3115$, mean $P_1 = 0.3335$, and mean $P(n \geq 2) = 0.3550$. The maximum P_1 observed in the run is 0.38175.

S4.1 Parity structure

The Hamiltonian terms $\hat{a}^\dagger \hat{a}$, $\hat{a}^{\dagger 2} \hat{a}^2$, and $\hat{a}^{\dagger 2} + \hat{a}^2$ commute with the number-parity operator $(-1)^n$. Starting from the vacuum, the unitary sector therefore populates only even Fock states. Odd populations arise through the single-quantum jump operator $\hat{a}$.

This establishes that P_1 is not coherently addressed by the pair Hamiltonian. The present calculation does not provide deterministic coherent preparation of $|1\rangle$. A rigorous universal upper bound on transient P_1 for every driven-dissipative parameter choice is not claimed here.

S4.2 Anomalous-correlator convention

Candidate reduced correlator	Complex RMS	Magnitude RMS	Magnitude correlation	Mean phase shift	Phase locking
uv	1.7600	0.4817	0.3411	-2.2474 rad	0.0199
$u v^*$	0.6702	0.4817	0.3411	0.0145 rad	0.8122

Candidate reduced correlator	Complex RMS	Magnitude RMS	Magnitude correlation	Mean phase shift	Phase locking
-uv	1.7535	0.4817	0.3411	0.8941 rad	0.0199
-u v*	2.3923	0.4817	0.3411	-3.1271 rad	0.8122

The convention $u v^*$ gives the best complex agreement with the Lindblad anomalous correlator $\langle \hat{a}\hat{a} \rangle$. Its near-zero mean phase offset and phase-locking resultant 0.812 indicate that the reduced model captures the pair-phase trend more reliably than the pair-correlation magnitude.

The magnitude RMS difference remains 0.482 and the magnitude correlation is 0.341. Therefore the reduced pair amplitude is only qualitative.

Supplement 5. Pair-phase control and dynamical step response

S5.1 Stationary transfer of the external drive phase

For the resonantly modulated protocol $G_\varphi(t) = \eta\Omega \sin(2\Omega t + \varphi_d)$, five Lindblad calculations were performed at $\varphi_d = 0, \pi/4, \pi/2, 3\pi/4$, and π . The late-time rotating-frame pair phase was defined as $\Phi_{\text{pair}}(t) = \arg\langle \hat{a}\hat{a} \rangle + 2\Omega t$. The fitted relation is

$$\bar{\Phi}_{\text{pair}} = -1.5589 - 0.99984 \varphi_d, \quad R^2 = 0.9999993,$$

with an RMS fit deviation of 9.3×10^{-4} rad. Thus the stationary pair phase follows the imposed pump phase nearly one-to-one, with the minus sign fixed by the adopted phase convention. Across the same phase range, the late-time mean occupation changes only from 1.2133 to 1.2252 and the mean pair-correlation amplitude changes from 1.1035 to 1.1309. The circular phase standard deviation remains 0.225-0.232 rad. These are temporal variations of the ensemble-averaged correlator, not a quantum-trajectory phase-diffusion measurement.

S5.2 Dynamical phase-step protocol

A phase step breaks the time-translation equivalence of the stationary five-phase scan and therefore tests genuine dynamical control. The Hamiltonian discontinuity at $\Omega t = 100$ is integrated as two separate mesolve segments: the final density matrix of the zero-phase segment is used as the initial state of the post-step segment, while absolute time is retained in the rotating-frame transformation.

The analysis uses the complex rotating-frame correlator $z(t) = \langle \hat{a}\hat{a} \rangle(t) e^{2i\Omega t}$. The response time is extracted from the complex distance $|z(t) - z_\infty|$, avoiding phase unwrapping when the amplitude passes through zero during large phase steps. This convention is documented in the archived script `phase_step_test_v2.py`.

For phase steps $\varphi_{\text{step}} = \pi/8, \pi/4, \pi/2$, and π , the steady-state transfer ratios $-\Delta\Phi/\varphi_{\text{step}}$ are 0.99477, 1.00045, 0.99764, and 0.99688, respectively. The corresponding complex-fit response times are 14.485, 19.957, 26.120, and 21.734 Ω^{-1} , all shorter than the damping time $(\alpha\Omega)^{-1} = 100 \Omega^{-1}$. The response is therefore drive-controlled rather than set by slow damping.

The phase-only interpretation has a finite operating range. For $\pi/8$ and $\pi/4$ steps the minimum pair-amplitude ratios during the transient are 0.703 and 0.451. For $\pi/2$ and π steps they fall to 0.005 and 0.015, so the pair correlator passes close to the origin and its instantaneous phase is temporarily ill-defined. The corresponding peak occupations rise to 2.168 and 2.219 times their pre-step values. Large phase steps therefore involve strong amplitude-phase coupling rather than a pure phase rotation.

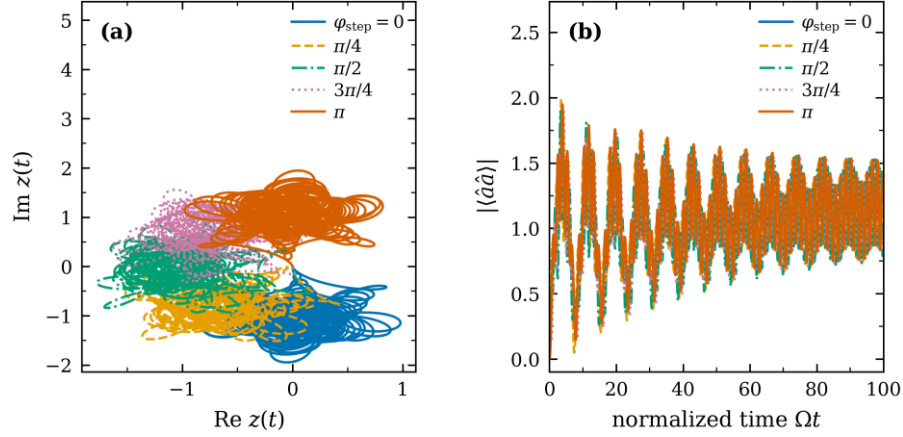


Figure S3. Complex-plane step response of the anomalous pair mode. The pump phase is switched at $\Omega t = 100$ from zero to $\varphi_{\text{step}} = 0, \pi/8, \pi/4, \pi/2$, and π . Left: trajectories of $z(t) = \langle \hat{a} \hat{a} \rangle \exp(2i\Omega t)$ in the complex plane. Right: pair-correlation amplitude across the step. The final phase transfer is nearly one-to-one for all tested steps. For $\pi/2$ and π , however, the pair amplitude collapses close to zero, so the instantaneous phase is temporarily ill-defined and phase and amplitude are not dynamically decoupled.

Supplement 6. Energy scale and architectural projections

The 22–24 aJ switching-event estimate is obtained from the experimentally reported absorbed-energy density multiplied by the assumed $20 \times 20 \times 10 \text{ nm}^3$ cell volume. It is not calculated from $\langle n \rangle$, $|v|^2$, $\hbar\Omega$, or the dimensionless Kerr coefficient U .

The normalized calculations demonstrate bounded dynamics for specified model protocols. They do not independently establish a microscopic mapping from U to a material-specific YIG:Co Kerr coefficient. Such a mapping would require the anisotropy, saturation magnetization, mode volume, and spatial mode profile.

Throughput, passive-cooling, readout-energy, fabrication-yield, and multi-cycle-logic figures require a separate device-level model and are not validated outputs of the present single-mode simulations. No such figures are claimed in this paper.

The software archive contains: the five-phase stationary test scripts, the phase-step response scripts, the publication-figure generator, the finite-temperature frequency-scan scripts and results (50 GHz–10 THz), CSV and JSON numerical summaries, and this Supplementary Information document.

All calculation scripts are archived at the project repository: <https://github.com/alexandermtishin-art/Nonlinear-magnon-dynamics-in-Co-doped-YIG-four-operating-regimes>. The archive contains the Python/QuTiP scripts for each figure, processed CSV outputs, and a SHA256 checksum manifest.

Supplement 7. Reproducibility checklist

1. Record the exact Hamiltonian, including the coefficient and convention of the Kerr term.
2. State whether the pair coefficient is static in the calculation frame or explicitly modulated in time.
3. Define η independently from any coefficient that is merely labelled by η .
4. Use $\kappa = 2\alpha\Omega$ when $\alpha\Omega$ is intended as the amplitude-decay rate.
5. Distinguish final-time values, late-time means, maxima, and time-averaged standard deviations.
6. Distinguish $|\nu|^2$, $\langle n \rangle$, P_1 , P_0+P_1 , and $P(n \geq 2)$.
7. Report basis-cutoff convergence separately from physical low-level confinement.
8. Archive the code version, package versions, random seeds if applicable, and all generated CSV files.
9. Do not infer a Fock-state fidelity from a mean occupation close to one.
10. Do not infer phase diffusion from a single ensemble-averaged Lindblad trajectory.

Supplement 8. Finite-temperature frequency scan at 310 K

S8.1 Model and numerical protocol

The room-temperature frequency scan uses the thermal Lindblad equation with the same single-mode operators as above, but with both downward and upward bath channels:

$$L\downarrow = [\kappa(\bar{n}_{\text{th}}+1)]^{1/2} \hat{a}, \quad L\uparrow = (\kappa\bar{n}_{\text{th}})^{1/2} \hat{a}^\dagger, \quad \bar{n}_{\text{th}} = [\exp(hf/k_B T) - 1]^{-1}.$$

The scan Hamiltonian is $H = \Omega n + U \hat{a}^\dagger \hat{a}^2 + (\eta\Omega/4)(\hat{a}^\dagger + \hat{a})^2$, with $\Omega = 1$, $\eta = 1.2$, $\alpha = 10^{-3}$, $\kappa = 2\alpha\Omega$, $T = 310$ K, a 140-state Fock basis, vacuum initial state, 20 normalized cycles, and 10 requested output points per cycle. Frequencies are 50, 75, 100, 150, 200, 300, 500, and 750 GHz, followed by 1, 1.5, 2, 3, 4, 5, 6, 7, 8, 9, and 10 THz.

Convention warning. The U used in this frequency scan multiplies $\hat{a}^\dagger \hat{a}^2$ directly. It is therefore not numerically interchangeable with the matched audit convention $(U\Omega/2)\hat{a}^\dagger \hat{a}^2$ used in Supplements 3–5. Results from the two conventions must remain explicitly labelled.

For each frequency the archived output records the final mean occupation $\langle n \rangle$, the low-manifold population P_0+P_1 , the leakage $P(n \geq 2)$, the anomalous pair amplitude $|\langle \hat{a}\hat{a} \rangle|$, the normalized pair measure $|\langle \hat{a}\hat{a} \rangle|/(2\langle n \rangle + 1)$, and the population of the ten highest basis levels.

S8.2 Completed 6–10 THz results for $U = 0.55$

The final high-frequency points of the completed scan are:

f (THz)	$\bar{n}_{\text{th}}$	Final $\langle n \rangle$	P_0+P_1	$P(n \geq 2)$	$ \langle \hat{a}\hat{a} \rangle $	Normalized pair
6	0.652875	0.245215	0.925418	0.074582	0.393312	0.263892
7	0.511353	0.228658	0.926351	0.073649	0.397639	0.272857
8	0.408083	0.216617	0.926733	0.073267	0.401108	0.279862
9	0.330225	0.207562	0.926841	0.073159	0.403908	0.285422
10	0.270071	0.200580	0.926811	0.073189	0.406184	0.289891

Across 6–10 THz the low-manifold population rises only from 0.9254 to 0.9268, while the final mean occupation falls from 0.2452 to 0.2006. The absolute pair amplitude rises slightly from 0.393 to 0.406 and the normalized pair measure increases from 0.1949 to 0.2164. The population in the highest ten basis levels is below 10^{-118} at every point, so the $N = 140$ truncation is not limiting these high-frequency results.

The scan does not show a sharp transition at 6 THz. Recovery of low-Fock confinement is progressive across the several-THz range: P_0+P_1 is 0.9079 at 3 THz, 0.9187 at 4 THz, 0.9233 at 5 THz, and 0.9254 at 6 THz. Accordingly, 6 THz should be treated as a representative room-temperature operating point or an engineering criterion, not as a mathematically resolved threshold. This frequency scan uses the static effective pair drive, whose Hamiltonian is time-independent, so the phase of $\langle \hat{a}\hat{a} \rangle$ relaxes to a constant and the scan does not define a within-run phase statistic: the circular standard deviations quoted in Section 6 and Table 3 of the main text come from separate runs of the resonantly modulated phase-control protocol and must not be read off this scan. What the scan does establish for pair-phase

encoding is the thermal input to those runs, the occupation $\bar{n}_{\text{th}}$ of the driven mode, which falls from 128.7 at 50 GHz to 0.65 at 6 THz.

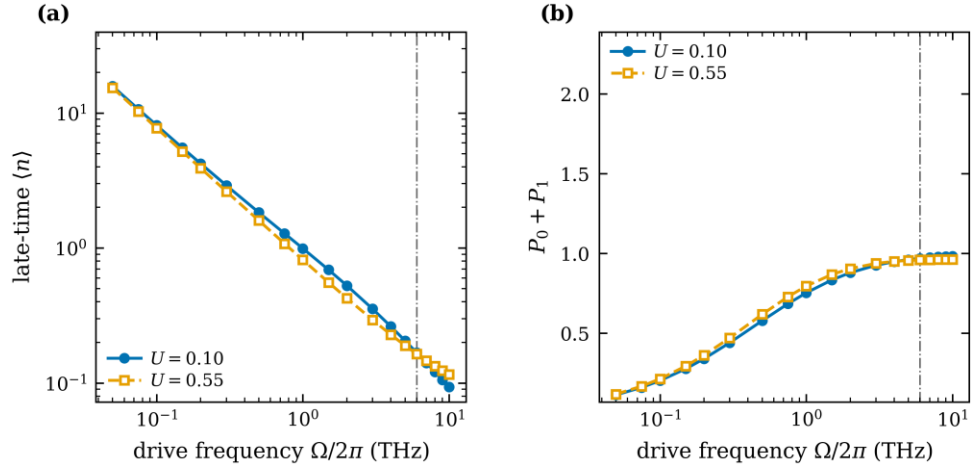


Figure S4. Finite-temperature frequency scan at 310 K. Final mean occupation and low-manifold population are shown for $U = 0.10$ and $U = 0.55$; leakage and pair-correlation measures are shown for $U = 0.55$. The frequency axis is logarithmic. The static effective pair-drive scan demonstrates progressive thermal recovery with frequency and no singular threshold at 6 THz.